

**Time:** 5 minutes**Outcome:** Connect Elasticity and Total Revenue.**Difficulty:** ●●○ Intermediate

Elasticity and Total Revenue



THE CORE IDEA

Total revenue equals price multiplied by quantity, and the effect of a price change depends on demand elasticity. A point on demand supplies the price and quantity needed to calculate revenue. At unit elasticity the price and quantity percentage effects offset, marking the revenue peak on a standard linear demand curve.



HOW TO RECOGNIZE IT

- Calculate total revenue as price multiplied by quantity at each relevant point.
- If price and total revenue move in opposite directions, demand is elastic over that range.
- If price and total revenue move in the same direction, demand is inelastic over that range.
- At unit elasticity, a small price change leaves total revenue approximately unchanged.



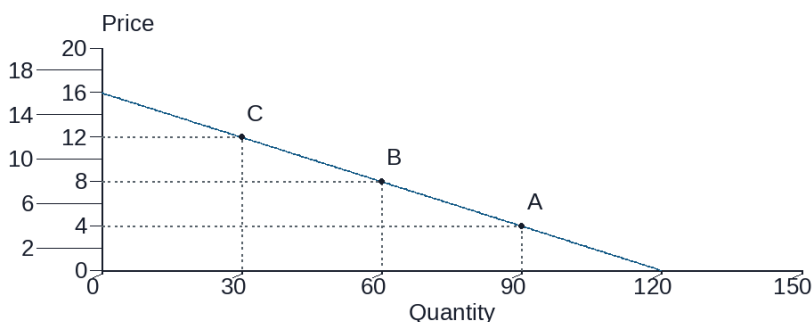
WATCH OUT

A common mistake is to compute elasticity without checking the associated total-revenue change. With elastic demand, price and total revenue move in opposite directions; with inelastic demand, they move together.



WORKED EXAMPLE: ELASTICITY AND REVENUE

— Demand



At point C, price is \$12 and quantity is 30, so total revenue is $\$12 \times 30 = \360 . At point B, price is \$8 and quantity is 60, so revenue is \$480. Moving from C to B lowers price but raises total revenue by \$120, indicating elastic demand over that segment.



CHECK YOURSELF

Cruise-cabin price falls from \$3,200 to \$2,880 while quantity rises from 1,500 to 1,900. Calculate midpoint elasticity and both revenue totals. Is demand elastic over this range, and does revenue rise or fall?

**READY?**

Return to the game and master this concept.